

A FIVE-VERTEX COUNTEREXAMPLE TO TANG'S BUNKBED SELF-AVOIDING-WALK QUESTION

ABSTRACT. Tang asked whether, for every finite connected simple graph G , the number of self-avoiding walks in $G \square K_2$ from u_0 to v_0 is at most the number from u_0 to v_1 whenever uv is not a cut edge. We answer this question negatively. For the five-vertex house graph and a cycle edge ad , exact enumeration gives

$$|\mathcal{S}(a_0, d_0)| = 60 > 59 = |\mathcal{S}(a_0, d_1)|.$$

The base graph is 2-connected, and five is the minimum possible order among counterexamples with an adjacent non-cut edge. We also give a 3-connected counterexample on seven vertices.

1. A MINIMUM-ORDER COUNTEREXAMPLE

For a finite connected simple graph G , write $x_i = (x, i)$ for a vertex of the bunkbed graph $G \square K_2$. For vertices p, q of the bunkbed graph, let $\mathcal{S}(p, q)$ denote the set of self-avoiding walks from p to q . Tang asked whether

$$(1) \quad |\mathcal{S}(u_0, v_0)| \leq |\mathcal{S}(u_0, v_1)|$$

holds whenever (u, v) is not a cut edge of G [1, Question 3.3]. The example below uses an adjacent pair, so it gives a negative answer independently of any convention concerning nonadjacent pairs.

Theorem 1. *Let*

$$V(G) = \{a, b, c, d, e\}, \quad E(G) = \{ab, ad, ae, bc, be, cd\}.$$

Then G is 2-connected, ad is not a cut edge, and

$$|\mathcal{S}(a_0, d_0)| = 60 > 59 = |\mathcal{S}(a_0, d_1)|.$$

Consequently, (1) is false. Moreover, no counterexample with an adjacent non-cut edge has fewer than five vertices.

Proof. The graph G is the house graph: it is the union of the cycle $abcd$ and the triangle abe . Deleting any vertex leaves a connected graph, so G is 2-connected. In particular, the cycle edge ad is not a cut edge.

We use an exact last-step recurrence. Let $B = G \square K_2$, fix $s \in V(B)$, and, for $A \subseteq V(B)$ and $x \in A$, let $F_s(A, x)$ be the number of self-avoiding walks from s to x whose vertex set is exactly A . Set

$$F_s(\{s\}, s) = 1, \quad F_s(A, s) = 0 \quad \text{for } A \neq \{s\},$$

and set $F_s(A, x) = 0$ if $s \notin A$. For $x \neq s$,

$$(2) \quad F_s(A, x) = \sum_{\substack{y \in A \setminus \{x\} \\ xy \in E(B)}} F_s(A \setminus \{x\}, y).$$

Indeed, deleting the final vertex x gives a bijection according to the unique penultimate vertex y . Hence, if $N_k(s, t)$ denotes the number of k -edge self-avoiding walks from s to t , then

$$(3) \quad N_k(s, t) = \sum_{\substack{A \subseteq V(B) \\ s, t \in A, |A|=k+1}} F_s(A, t).$$

Exact evaluation of (2)–(3) gives

k	1	2	3	4	5	6	7	8	9	total
$N_k(a_0, d_0)$	1	0	2	1	9	14	13	13	7	60
$N_k(a_0, d_1)$	0	2	0	7	9	12	16	9	4	59

and therefore gives the claimed strict reverse inequality. All computations use integer arithmetic.

It remains to verify minimum order. For an adjacent non-cut edge uv , define

$$\Delta_G(u, v) = |\mathcal{S}(u_0, v_0)| - |\mathcal{S}(u_0, v_1)|.$$

Up to isomorphism, the connected simple graphs on two, three, or four vertices are exactly those listed below. The final column is the maximum of $\Delta_G(u, v)$ over adjacent non-cut edges, evaluated by the same recurrence; “none” means that every edge is a cut edge.

$ V(G) $	G	$\max \Delta_G(u, v)$
2	K_2	none
3	P_3	none
3	K_3	−1
4	$P_4, K_{1,3}$	none
4	C_4	−1
4	paw graph	0
4	$K_4 - e$	−4
4	K_4	−14

The one-vertex case is vacuous. Thus no graph of order at most four violates (1) at an adjacent non-cut edge. $\square$

2. A 3-CONNECTED COUNTEREXAMPLE

The failure is not removed by imposing 3-connectivity.

Proposition 2. *Let H be the graph in which $\{0, 1, 2, 6\}$ induces K_4 , $\{3, 4, 5\}$ induces K_3 , and the remaining edges are 05, 14, and 23. Then H is 3-connected and*

$$|\mathcal{S}(0_0, 5_0)| = 5031 > 4801 = |\mathcal{S}(0_0, 5_1)|.$$

Proof. Deleting at most two vertices leaves each clique nonempty and connected. The three edges 05, 14, and 23 are vertex-disjoint, so at least one survives and joins the two remaining cliques. Hence H remains connected after deleting any two vertices, and therefore $\kappa(H) \geq 3$. Since $\delta(H) = 3$, we have $\kappa(H) = 3$. The exact counts follow from (2)–(3). $\square$

Remark 3. The examples above do not address Tang’s complete-graph conjecture [1, Conjecture 1.8].

REFERENCES

- [1] P. Tang, *Maximum flow and self-avoiding walk on bunkbed graphs*, arXiv:2502.06237v1 [math.PR], 2025.